

Evaluate the related series of each sequence.

11) 3.7, 4.6, 5.5, 6.4

12) 26, 36, 46, 56, 66, 76, 86

Evaluate each arithmetic series described.

13) $a_1 = \frac{3}{2}, d = \frac{1}{2}, n = 50$

14) $a_1 = -22, d = -3, n = 12$

15) $a_1 = 20, a_n = 55, n = 6$

16) $a_1 = 14, a_n = 86, n = 10$

17) $\sum_{m=1}^{10} \left(-2 + \frac{4}{3}m \right)$

18) $\sum_{n=1}^9 (7n - 17)$

19) $\sum_{m=1}^{35} (7m - 12)$

20) $\sum_{k=1}^{14} \left(-\frac{2}{3} + \frac{1}{2}k \right)$

21) $\sum_{i=2}^{10} (0.8i - 6.3)$

22) $\sum_{n=4}^{53} (2n - 7)$

Determine the number of terms n in each arithmetic series.

23) $1 + (-2) + (-5) + (-8) \dots, S_n = -259$

24) $18 + 21 + 24 + 27 \dots, S_n = 468$